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Risk Management Group Work Project 1

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On AI Usage

This research made use of artificial intelligence tools in three specific capacities: (i) to assist with literature search and verification of academic sources and datasets; (ii) to help refine and optimize algorithmic implementations during the structure learning phase; and (iii) for grammar, spelling, and style corrections, particularly for itemized content where precision in phrasing mattered.

All intellectual contributions, theoretical framing, empirical analysis, and final interpretations are entirely our own. The AI tools served only as supporting aids, not as co-authors, and we assume full responsibility for any errors or omissions.

1 Problem Formulation

The global crude oil market is among the most complex and closely watched commodity systems out there. Unlike your typical financial equities, crude oil's spot price isn't just about discounting expected future cash flows—it acts more like a vital macroeconomic pulse, shaping industrial costs, swaying inflation, and even nudging geopolitical decisions. So, trying to forecast crude oil prices isn't really a straightforward time-series exercise. What's needed is a broader framework, one that can tease apart how macroeconomic forces, supply-chain mechanics at the micro level, and unexpected geopolitical events all intertwine.

1.1 Limitations of Traditional Econometric Approaches

Looking back at how quantitative analysts and economists have approached this problem, they've tended to lean on familiar workhorses like ARIMA and GARCH models. There's no denying these are mathematically solid and they do a decent job picking up things like volatility clustering or mean reversion. But if we apply them to today's oil market — which is driven by so many interconnected variables — they start to show their weaknesses.

One big issue is that these models are fundamentally atheoretical. They're good at finding statistical correlations but they don't really tell you anything about causation. The GARCH model, for example, it might predict that volatility will carry on being volatile, but it can't explain *why* a supply disruption in the Middle East ends up affecting refinery margins in the U.S. The causal pathway just isn't there.

Another problem is how they handle data at different frequencies and structural breaks. If we get macroeconomic data like GDP or CPI coming out monthly or quarterly, while futures prices are updating every second. Mixing those together is messy. And then you've got these sudden regime shifts, the 2014 shale revolution, or the demand collapse during COVID, which completely break the stationarity assumptions that these classical models depend on.

Plus, we can see that these models don't handle missing data well at all. And when we put añ together global microeconomic data, especially from countries that aren't exactly transparent about their inventory levels, missing values are pretty much the norm.

1.2 The Suitability of Probabilistic Graphical Models

To get around these limitations, the proposal is a Probabilistic Graphical Models (PGMs), specifically Belief Networks (Bayesian Networks). PGMs give us a elegant way to combine graph theory with probability theory, which lets us represent complex, high-dimensional joint distributions in a compact form.

A Belief Network is basically a Directed Acyclic Graph (DAG). The nodes are random variables, as OPEC production quotas, interest rates from the Fed, or the Brent-WTI spread, and the directed edges show conditional dependencies and causal influences. What makes this architecture so powerful is the local Markov property: once you condition on its parents, a node becomes independent of all its non-descendants. This drastically cuts down the computational complexity, turning what would be an intractable joint distribution into a product of much simpler conditional probability distributions.

1.3 Advantages of the Proposed Methodology

There are several concrete advantages to using Belief Networks for forecasting crude oil prices, especially when we look at it through the three lenses of macroeconomic, microeconomic, and financial specialisation.

1.3.1 Advantages of the Proposed Methodology

Interpretability: Unlike deep neural networks or other black-box approaches, PGMs give us transparent causal pathways that you can actually trace and understand. Through structure learning, the model can spot hidden dependencies in the data that might either confirm or challenge existing economic theories. For someone managing a portfolio or making policy decisions, being able to visualise exactly how a shock to the US Dollar Index flows through to oil demand in emerging markets, that's invaluable for running scenarios and stress tests.

handling of uncertainty and missing data. Because Belief Networks are probabilistic by nature, they keep working even when some variables aren't observed. Say weekly crude stock data is late or just not available, the network can still marginalise over that uncertainty and give you a probability distribution for the spot price. That's way more useful than a single point estimate that might be completely off.

Blend expert knowledge with data-driven learning: The algorithm figure out the structure from the data, but also impose constraints based on what we know about the world. For example, it might be possible to specify that geopolitical risk can influence supply, but actually the supply can't change past geopolitical events. That kind of guidance helps the model avoid picking up spurious correlations that don't make economic sense.

Regime-switching angle. Combining PGMs with latent variable models like Hidden Markov Models, this framework could be adapted to a different market conditions on the fly, whether we're in contango or backwardation, or during high-volatility crisis periods versus calmer growth phases. Energy markets are notoriously non-linear, and this gives us a way to capture those changing dynamics.

So overall, the idea is to build something that's not just statistically sophisticated but also causally informed and robust to the kind of messiness you inevitably run into with real-world data.

To sum up, the core aim here is to move beyond purely statistical time-series forecasting, building a framework that's causally aware and probabilistically robust. By learning both

the structure and parameters of a Belief Network from a dataset that spans macroeconomic, microeconomic, and financial domains.

2 Data Identification, Sources and Perspectives

We broke down the data into three distinct angles: the macroeconomic and geopolitical side, the microeconomic side, and the financial side. Each one brings something different to the table.

2.1 Macroeconomic and Geopolitical Perspective

This lens is all about the big picture: sovereign policies, global systemic drivers, that sort of thing. We're looking at variables like the Consumer Price Index (CPI) and the Federal Funds Rate, which provide a sense of the broader inflationary environment and the cost of capital. Those are the kinds of things that shape global aggregate demand. Industrial Production is another key one; it's a pretty direct measure of real economic activity and how much energy the manufacturing sector is consuming. We've also included the US Dollar Index, and that's because crude oil is priced in dollars globally. So when the dollar's purchasing power shifts, it feeds back into nominal oil prices, usually in the opposite direction, and changes how demand behaves in emerging markets.

2.2 Microeconomic Perspective

Trying to get high-frequency microeconomic data, like individual refinery margins or real-time rig counts is a challenge by itself. Reporting lags, proprietary restrictions, it's a whole thing. But even without that granular level of detail, we can still capture the physical market balance through the WTI spot price and the futures curves. The relationship between commercial inventory levels and extraction costs tends to shape short-term physical supply responses, and that ultimately shows up in the volatility of spot and near-month futures contracts.

2.3 Financial Perspective

We're using front-month WTI and Brent crude futures contracts to get at market sentiment, where speculative positions are sitting, and the geographical arbitrage play, that's the Brent-WTI spread. These instruments reflect how the market prices risk at the margin, and they tend to react pretty sharply to macroeconomic news and geopolitical developments.

3 Data Dictionary and Summary Table

We pulled the raw data from two main sources. For the high-frequency market stuff, we went with Yahoo Finance, because it's reliable and gives us good coverage. For the macroeconomic indicators, we used the Federal Reserve Economic Data (FRED) repository, which is pretty much the gold standard for that kind of thing.

Now, one challenge was getting everything to line up properly. The market data comes in at daily frequency, but the macro stuff is monthly. So we aligned all the daily observations to a monthly frequency using the Month-End convention (that's the ME approach). This way everything syncs up nicely with the macroeconomic releases. This resulted in a balanced panel of 199 monthly observations, covering January 2010 right through to the present.

Table 1 below gives a full dictionary of all the variables we've integrated into the unified dataset.

Table 1: Data Dictionary and Summary of Key Variables

Variable (Ticker)	Description	Native Freq.	Source	Date Range
WTI_CL	WTI Crude Oil Futures (Front Month)	Daily	Yahoo Finance	2010-01 to Present
Brent_BZ	Brent Crude Oil Futures (Front Month)	Daily	Yahoo Finance	2010-01 to Present
USD_Idx	US Dollar Index (Basket of Currencies)	Daily	Yahoo Finance	2010-01 to Present
CPI	Consumer Price Index for All Urban Consumers	Monthly	FRED (CPIAUCSL)	2010-01 to Present
Fed_Funds	Effective Federal Funds Rate	Monthly	FRED (FEDFUNDS)	2010-01 to Present
Ind_Prod	Industrial Production Index	Monthly	FRED (INDPRO)	2010-01 to Present
WTI_Spot	Spot Price of WTI Crude Oil	Daily	FRED (DCOILWTICO)	2010-01 to Present

4 Data Preprocessing and Cleaning

So we followed a sterilisation protocol of a three-stage cleaning process:

1. Extreme Outlier Identification and Clipping
2. Bad Data and Duplicated Indices
3. Missing Values and Temporal Interpolation

4.1 Extreme Outlier Identification and Clipping

We went with the Interquartile Range (IQR) method. The bounds were set at $Q1 - 3 \times IQR$ and $Q3 + 3 \times IQR$. We used factor of 3 used for standard boxplots because commodity returns are inherently leptokurtic, they have these fat tails where extreme price movements, while statistically rare, are actually economically meaningful. Considering for example the 2008 financial crisis or the COVID demand shock in 2020; those weren't outliers in the sense of "bad data," they were real events.

When we ran the preprocessing pipeline, only two extreme values got flagged and clipped across the whole dataset. That's a pretty low number, which tells us the FRED and Yahoo Finance feeds are good quality. The values we clipped were most likely flash-crash anomalies or erroneous ticks that the exchanges later corrected, not genuine structural market events.

4.2 Bad Data and Duplicated Indices

We aligned everything to the last business day of each month, then the chances of duplicated chronological indices were already pretty low. We confirm with zero duplicated indices in both the raw and aggregated dataframes, so we ended up with a strictly monotonic time-series progression. No issues there.

4.3 Missing Values and Temporal Interpolation

We used time-based interpolation for to go with mean or median imputation. By this approach we aimed to destroy the temporal autocorrelation structure, which is kind of a big deal for financial time series. Time-based interpolation, on the other hand, respects the chronological distance between known data points. For any remaining missing values at the very beginning or end of the dataset, we used forward-filling and backward-filling. After all this, we had a complete, balanced panel dataset with no null values, ready for exploratory analysis and probabilistic modelling.

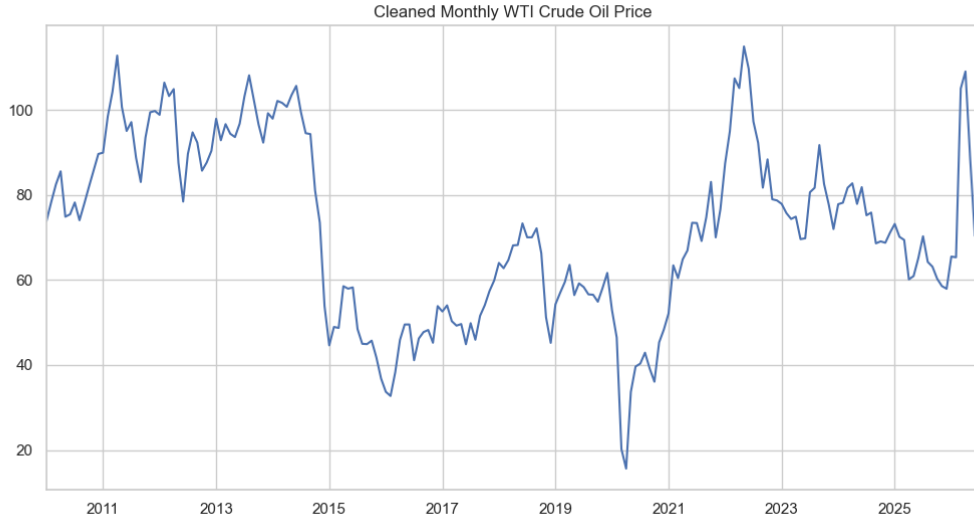


Figure 1: Sterilised WTI Crude Oil Futures Price (2010–Present). The blue line represents the robust, outlier-clipped series, demonstrating the preservation of major structural breaks such as the 2014 shale glut and the 2020 pandemic crash.

5 Exploratory Data Analysis

Once we had the sterilised monthly dataset put together, we dove into a thorough Exploratory Data Analysis (EDA) in order to the underlying statistical properties of the crude oil market. We broke it down into distributional, temporal, and multivariate dimensions, and by the end of it, we'd identified the key stylised facts that tend to characterise energy commodities.

5.1 Distributional and Temporal Analysis

Figure 2 a non-Gaussian. The monthly percentage returns have this pronounced leptokurtic shape. This is, there's a sharp peak around the mean, but then these heavy, asymmetric tails on either side. So oil prices can cruise along with relative stability for months at a time, but they're also prone to these sudden, violent shocks in either direction.

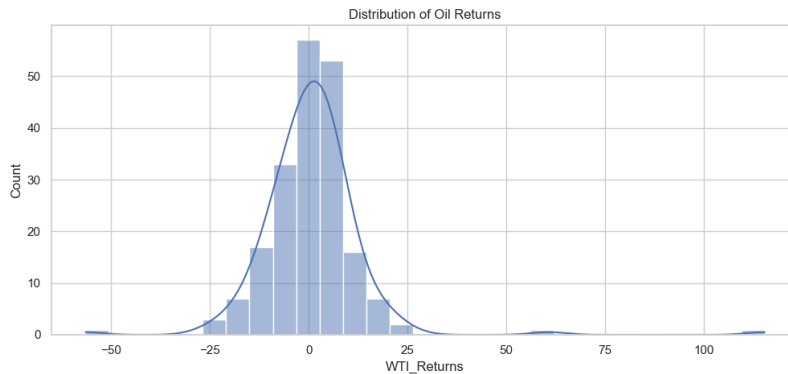


Figure 2: Distribution of Monthly WTI Percentage Returns. The pronounced kurtosis and fat tails demonstrate the failure of normal distribution assumptions in energy markets.

The time-series plot of the sterilised WTI price (Figure 1) tells an interesting story: several

distinct market regimes playing out. From 2010 to 2014, prices were consistently high, fuelled by strong demand from emerging markets. Then came the structural break in late 2014, that's when the US shale revolution really kicked in, leading to a supply glut. And looking at early 2020, that's where things get really wild. Unprecedented demand destruction met storage capacity constraints, and futures prices briefly went negative. Something not very common.

5.2 Multivariate Dependencies

To get a sense of how the macroeconomic, microeconomic, and financial variables interact with each other, we generated a correlation matrix of the monthly percentage returns.

Figure 3 confirms what economic theory would predict. The most obvious relationship is the persistent negative correlation between the US Dollar Index and both WTI and Brent returns. Makes sense when considering oil's priced in dollars, so when the dollar strengthens, oil becomes more expensive for anyone holding foreign currency. That tends to dampen demand and push prices down. On the flip side, industrial production and inflation metrics generally show positive correlations with energy prices, though there's usually a lag. That reflects the derived demand for crude

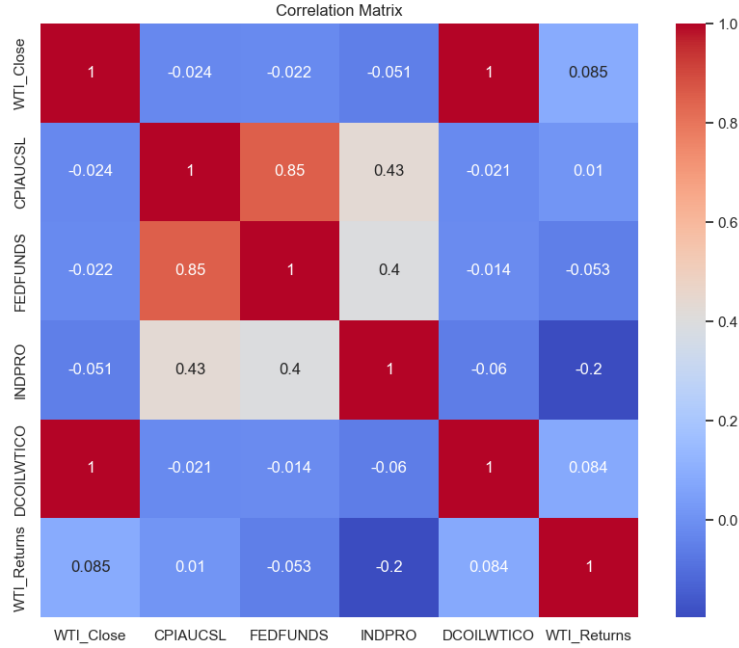


Figure 3: Correlation Heatmap of Monthly Returns. Notable inverse relationships are observed between the US Dollar Index and crude oil returns, validating the financial transmission mechanism of currency fluctuations.

5.3 Stylised Facts of Oil Prices

Based on everything we saw in the EDA, we can pin down some stylised facts about crude oil prices. Any forecasting model worth its salt, including our proposed Belief Network — needs to be able to capture these:

1. **Volatility Clustering and Structural Breaks:** Unlike standard equity markets, oil prices are highly susceptible to exogenous supply shocks and geopolitical events. This

results in distinct periods of extreme volatility clustering, separated by structural breaks where the underlying data-generating process fundamentally shifts (e.g., the transition from a supply-constrained market to a demand-constrained market).

2. **Non-Normal Return Distributions:** Oil returns are distinctly non-Gaussian. They exhibit high kurtosis (fat tails) and skewness, meaning that extreme events (e.g., a $> 20\%$ monthly move) occur far more frequently than a normal distribution would predict.
3. **Weak Autocorrelation in Returns, Strong Persistence in Volatility:** The raw monthly returns show very low linear autocorrelation, suggesting that the market is highly efficient and past price directions do not easily predict future directions. However, the squared returns (a proxy for volatility) exhibit strong positive autocorrelation. This confirms that while the direction of the price change is unpredictable, the magnitude of the change is highly persistent.
4. **Macroeconomic and Financial Sensitivity:** Oil prices do not exist in a vacuum. They are deeply sensitive to the cost of capital (interest rates), global industrial activity, and currency fluctuations (USD strength). The causal propagation of these variables is non-linear and regime-dependent, necessitating a modelling framework capable of handling complex conditional dependencies.

6 Probabilistic Graphical Models

Looking at everything through those three lenses we've been using macroeconomic, microeconomic, and financial. The goal here is to lay down the mathematical and conceptual foundations we need to model the complex, interdependent dynamics of the global crude oil market.

6.1 Macroeconomic and Geopolitical Perspective: Foundations of Probabilistic Graphical Models

Looking at the global oil market from a macroeconomic and geopolitical angle, it's really just a giant web of interdependent variables. Sovereign policies, aggregate demand, geopolitical shocks, they all propagate through these intricate causal pathways. Probabilistic Graphical Models (PGMs) give us a way to represent that kind of system mathematically. They combine graph theory with probability theory, letting us compactly represent a joint probability distribution over a high-dimensional set of random variables. It maps variables to nodes, probabilistic dependencies to edges for visualising and compute how macroeconomic shocks cascade through the system.

PGMs generally fall into two broad families: Belief Networks and Markov Networks.

- **Belief Networks (Bayesian Networks):** These are directed acyclic graphs (DAGs). The directed edges represent explicit causal influence or temporal precedence. Each node comes with a conditional probability distribution that quantifies the effect of its parent nodes. For oil markets, Belief Networks are particularly well suited to modelling asymmetric causal chains. This can be said: a geopolitical conflict (parent) disrupts supply chains (child), which then drives up inflation (grandchild). The acyclic nature enforces a logical flow of time and causality, which makes sense for these kinds of problems.
- **Markov Networks (Markov Random Fields):** These use undirected graphs and are parameterised by potential functions over cliques of nodes, rather than conditional probabilities. They're really effective for systems with simultaneous, bidirectional feedback loops, situations where causal directionality isn't clear. For example, the mutual influence between global industrial production and aggregate energy consumption often happens

simultaneously within a given fiscal quarter. Undirected representations are better at capturing those equilibrium states.

For forecasting crude oil prices, the goal is primarily leveraging Belief Networks. Being able to encode directional causal influence and perform both predictive (forward) and diagnostic (backward) reasoning is pretty essential for macroeconomic scenario analysis.

6.2 Microeconomic Perspective: Parameter Learning versus Structure Learning

From microeconomic lens, its possible to look at individual actors, refinery utilisation rates, inventory drawdowns, upstream capital expenditure, that sort of thing. To capture these dynamics accurately within a PGM, that we need to deal with in two distinct phases of model construction: parameter learning and structure learning.

Parameter Learning assumes you already know the structural topology of the graphical model, the DAG, maybe from established microeconomic theory or expert consensus. Then it's just a matter of estimating the numerical parameters of the conditional probability distributions from the observed data. This is done, typically with Maximum Likelihood Estimation (MLE) or Bayesian estimation techniques. Say economic theory tells that commercial inventory levels directly influence short-term spot price volatility. Parameter learning will quantify the exact magnitude and variance of that relationship based on historical microeconomic data.

Structure Learning, This is about discovering the underlying graphical topology directly from the data. A hard computational problem due the space of possible DAGs grows super-exponentially with the number of variables. Structure learning algorithms generally come in two flavours: constraint-based methods (which use statistical conditional independence tests to eliminate edges) and score-based methods (which search the graph space to optimise a goodness-of-fit metric penalised for complexity, like the Bayesian Information Criterion). For the crude oil market, structure learning is incredibly valuable. It lets the model autonomously uncover hidden, non-linear microeconomic dependencies that human analysts might miss, reducing reliance on potentially biased expert input.

6.3 Financial Specialist Perspective: Markov Chains and Markov Blankets

From a financial specialist's perspective, crude oil is volatile asset class. Modelling its temporal evolution and feature space requires specific probabilistic tools, namely Markov Chains and the concept of the Markov Blanket.

Markov Chains and Hidden Markov Models (HMMs): A Markov Chain is a stochastic model where the probability of each event depends only on the state attained in the previous event. That's the memoryless property. Now, raw oil prices don't strictly follow a simple Markov Chain because there are long-term memory effects. But the *regimes* of the oil market often do follow one. Hidden Markov Models extend this by positing that the observed financial time series (say, daily returns) is generated by an underlying, unobservable Markov process representing distinct market states. So an HMM can dynamically classify the market into latent regimes, contango/low-volatility, backwardation/high-volatility, that kind of thing. This lets the forecasting model adapt its parameters dynamically based on the current financial regime.

The Markov Blanket: In a Belief Network, the Markov Blanket of a node X is the minimal set of variables that completely shields X from the rest of the network. Specifically, it consists of X 's parents, its children, and the other parents of its children (co-parents or spouses). The key property is that, conditional on this set, X is conditionally independent of all other nodes. For a financial specialist building a forecasting model for the WTI spot price, identifying the Markov Blanket of the price node is essentially optimal feature selection. It isolates the exact subset of macroeconomic, microeconomic, and financial indicators that have genuine predictive power, while discarding redundant variables that just add noise to the trading algorithm.

7 Structure Learning: The Inferred Causality Algorithm

The structure of the Belief Network has to be learned from our sterilised dataset, and that's where the Inferred Causality algorithm comes in. It's a constraint-based approach, rigorous, systematic, designed to uncover the causal DAG by methodically evaluating conditional independencies.

The algorithm works through three sequential phases, and it leans heavily on that Markov Blanket concept. First, it identifies the Markov Blanket for every variable, which drastically cuts down the dimensionality of the search space. Second, it isolates the direct neighbours (parents and children) by filtering out the spouses. Finally, it orients the undirected edges to establish causal directionality, with special attention paid to colliders, those v-structures that tell us a lot about causality.

The formal pseudocode is laid out in Algorithm 1 below.

Algorithm 1 Inferred Causality Structure Learning

Require: A dataset $\mathcal{D}$ containing N variables $\mathbf{V} = \{X_1, X_2, \dots, X_N\}$, and a significance level α for conditional independence tests.

Ensure: A Directed Acyclic Graph (DAG) $\mathcal{G} = (\mathbf{V}, E)$.

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1: Phase 1: Markov Blanket Discovery
2: for each variable  $X_i \in \mathbf{V}$  do
3:   Identify the Markov Blanket  $\mathcal{MB}(X_i)$  using a forward-backward selection algorithm
   (e.g., Grow-Shrink or IAMB).
4:   Enforce symmetry: If  $X_j \in \mathcal{MB}(X_i)$  but  $X_i \notin \mathcal{MB}(X_j)$ , remove the asymmetric de-
   pendency based on statistical reliability.
5: end for
6: Phase 2: Neighbourhood Identification (Undirected Skeleton)
7: for each variable  $X_i \in \mathbf{V}$  do
8:   Initialise the set of Parents and Children  $\mathcal{PC}(X_i) = \mathcal{MB}(X_i)$ .
9:   for each  $X_j \in \mathcal{PC}(X_i)$  do
10:    Search for a separating set  $\mathbf{S} \subseteq \mathcal{MB}(X_i) \setminus \{X_j\}$  such that  $X_i \perp X_j \mid \mathbf{S}$ .
11:    if such a separating set  $\mathbf{S}$  is found then
12:      Remove  $X_j$  from  $\mathcal{PC}(X_i)$  (identify  $X_j$  as a spouse/co-parent).
13:      Record  $\mathbf{S}$  in the separation set table  $\mathbf{Sep}(X_i, X_j)$ .
14:    end if
15:   end for
16: end for
17: Construct the undirected skeleton graph  $\mathcal{G}_{skel}$  using the validated  $\mathcal{PC}$  sets.
18: Phase 3: Edge Orientation (V-Structures and Meek's Rules)
19: for each unshielded triple  $X_i - X_k - X_j$  in  $\mathcal{G}_{skel}$  do
20:   if  $X_k \notin \mathbf{Sep}(X_i, X_j)$  then
21:     Orient as a collider (v-structure):  $X_i \rightarrow X_k \leftarrow X_j$ .
22:   end if
23: end for
24: Apply Meek's propagation rules to orient remaining edges without creating directed cycles
   or new v-structures.
25: return The final oriented DAG  $\mathcal{G}$ .

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